Aryan Bhagat

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EDUCATION

California State University, East Bay — **Aug 2024 – May 2026**
*Master of Science - **Computer Science**, GPA - 3.9/4.0* — *Hayward, CA, USA*

L.D. College of Engineering, Gujarat Technological University — **Oct 2020 – May 2024**
*Bachelor of Engineering - **Computer & Electronics Engineering**, CGPA - 8.8/10.0* — *Ahmedabad, India*

WORK EXPERIENCE

Teaching Associate — **Jul 2025 – May 2026**
College of Science, California State University - East Bay — *Hayward, CA, USA*

- Led **C++ Data Structures & OOP** (CS201) lab instruction for **30+ students**, driving mastery of **abstract data types, algorithm design**, and **time–space complexity (Big-O)** using STL, with a focus on writing efficient, scalable code.
- Improved **code quality** and **performance** by mentoring students in **debugging, profiling,** and **optimization**, guiding the implementation of high-performance **sorting** and **searching algorithms** and reinforcing software engineering best practices.

Research Assistant — **Sep 2025 – Mar 2026**
College of Science, California State University - East Bay — *Hayward, CA, USA*

- Built real-time perception pipelines on **JetAuto Pro** (Jetson Orin Nano) using **ROS2** and **3D depth cameras**, delivering accurate **gesture recognition, object detection,** and **autonomous path following** using Deep Learning and OpenCV.
- Evaluated AI-driven robotic systems with **AR-based visualization** for sign-language recognition, leveraging **LiDAR-based spatial mapping** and **sensor fusion** to improve navigation accuracy and reduce end-to-end system latency by 15%.

AI Engineer Intern — **Jul 2025 – Sep 2025**
Simple Ticket — *Boston, MA, USA*

- Engineered scalable **CRM chatbot systems** with **OCR-based document ingestion**, semantic knowledge grounding, and Retrieval-Augmented Generation (RAG) pipelines using **vector databases** and **LLMs**, orchestrated via LangGraph.
- Deployed production-grade, API-driven agent architectures using **RESTful services**, enabling dynamic agent orchestration, and multi-step reasoning pipelines with **LangChain Expression Language (LCEL)** and **stateful execution graphs**.

Firmware Engineer — **Jan 2024 – Jun 2024**
Indiesemic Private Limited — *Ahmedabad, India*

- Engineered embedded applications using **Zephyr RTOS** on **Nordic nRF52/nRF53 DKs** and **Quectel EVKs**, performing in-depth debugging with Segger J-Link, RTT, and Nordic nRF Connect SDK to optimize memory and CPU utilization.
- Architected a **BLE-based** data transmission framework with multi-protocol support **(UART, SPI, I2C, LoRa)**, enabling concurrent peripheral communication, reducing data latency, and improving overall system throughput by **10%**.

PROJECTS

Scalable Notification System [GitHub]

- Architected a **Kafka-based distributed notification pipeline** using priority-isolated topics and consumer groups, enabling parallel processing across **3 priority tiers and multi-channel delivery** (SendGrid for email, Twilio for SMS, Push).
- Built an **event-driven microservices architecture** with Spring Boot, integrating external APIs (SendGrid, Twilio) and implementing **asynchronous processing and priority-based routing** to improve delivery latency and system reliability.

AI Recipe Generation Platform (Smart Recipe Tagger) [GitHub]

- Crafted a full-stack AI platform using **React, Node.js, MongoDB,** and **Firebase**, integrating **Google OAuth** and **Google Photos API** to ingest and manage user images, with an analytics dashboard supporting scalable, real-time recipe exploration.
- Designed an **end-to-end multimodal AI pipeline** leveraging **Google Vision API** (ingredient detection) and **Gemini API** (recipe, nutrition, instruction generation), deployed via **Docker + Google Cloud Run** to achieve low-latency, scalable inference for automated recipe generation from images.

Cold Email Generator [GitHub]

- Developed an LLM-based tool using **Llama 3.3, Groq, LangChain, ChromaDB,** and **Streamlit** to generate personalized cold emails, focusing on improving relevance and reducing manual effort.
- Implemented data processing pipelines for **web scraping, text embedding,** and **vector similarity search**, enabling retrieval of relevant company and role context to ground LLM outputs, contributing to an 18% improvement in response rates.

Breast Cancer Classification using Supervised & Unsupervised Learning [GitHub]

- Implemented unsupervised learning pipelines using **spectral clustering** and **K-means**, with **majority-voting cluster** labeling, achieving 88% F1-score and validating model stability through **Monte Carlo simulations**.
- Developed and optimized a supervised classification model (**SVM with L1 regularization**), applying **5-fold cross-validation** to ensure generalization, achieving 94% precision, 92% recall, and 93% F1-score on breast cancer prediction tasks.

SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, Go, SQL
Backend & APIs: Node.js, Express.js, FastAPI, Flask, RESTful APIs, WebSockets
Frontend: React.js, Next.js, Vue.js, HTML5, CSS3, TailwindCSS, Chakra UI, Bootstrap, Framer Motion
ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, Hugging Face, Transformers, MLflow, RAG, Prompt Engineering
Databases & DevOps: PostgreSQL, MongoDB, NoSQL, ChromaDB, AWS (EC2, S3), Docker, CI/CD Pipelines, Git, GitHub
Fundamentals: DSA, OOP, Design Patterns, Debugging, Profiling, Performance Optimization, Agile, Testing